CDA: Post-hoc Domain Generalization by Distributional Canalization

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Setting: source-only post-hoc domain generalization. The training run is fixed, a checkpoint bank already exists, and deployment must choose a retained family and averaging weights without target labels.

Problem and Motivation

Domain generalization asks a model to learn from labeled source domains and generalize to an unseen target domain without target labels. In the post-hoc setting, the practical question is narrower and cleaner:

Given a checkpoint bank $\mathcal{B} = \{\theta_1, \dots, \theta_K\}$, which checkpoints should we average, and why?

Why existing post-hoc rules are incomplete

- Uniform late-window soups trust temporal adjacency.
- SWAD explains gains through flatter averaged solutions.
- DiWA and EoA explain gains through diversity and ensemble-style selection.
- None of these directly measures how fragile the deployed model is to *source-mixture shift*.

CDA thesis

- The deployment object should have low source fit, low source-mixture sensitivity, and low soup mismatch.
- We call this deployment principle *distributional canalization*.

Target-Risk Certificate

Let the source-domain loss vector be

$$L(\theta) = (L_1(\theta), \dots, L_E(\theta))^\top \in \mathbb{R}^E, \qquad \bar{L}(\theta) = \frac{1}{E}\mathbf{1}^\top L(\theta),$$

and let

$$P_\perp = I - \frac{1}{E}\mathbf{1}\mathbf{1}^\top$$

project onto the zero-sum subspace.
Under a local source-supported hidden-shift assumption, the target loss satisfies

$$L_T(\theta) \le \bar{L}(\theta) + \rho \, \|P_\perp L(\theta)\|_2 + \epsilon_{\text{app}}.$$

Interpretation

- $\bar{L}(\theta)$: average source fit
- $\|P_\perp L(\theta)\|_2$: sensitivity to redistributing source-domain mass
- ϵ_{app}: approximation gap of the local hidden-shift model

For soups, a mergeability term bridges endpoint-domain quantities to the deployed averaged model:

$$L_T(\bar{\theta}(w)) \le \bar{z}(w) + \rho \, \|P_\perp z(w)\|_2 + \left(E^{-1/2} + \rho\right) M(w) + \epsilon_{\text{app}}.$$

CDA Pipeline

Step 1: Version-space compression. For each source-support example (x_i, y_i), CDA tracks whether the checkpoint bank disagrees about the true-class confidence threshold:

$$D_{\mathcal{B}}(i;\gamma) = \mathbf{1}\left\{\exists j : q_{j,i} \ge \gamma \text{ and } \exists j' : q_{j',i} < \gamma\right\}.$$

$$d_{\text{vsc}}(S, \mathcal{B}; \gamma) = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\left\{D_S(i;\gamma) \ne D_{\mathcal{B}}(i;\gamma)\right\}.$$

This keeps a **non-contiguous** checkpoint family only when it still preserves the source-support disagreement structure that matters for deployment.

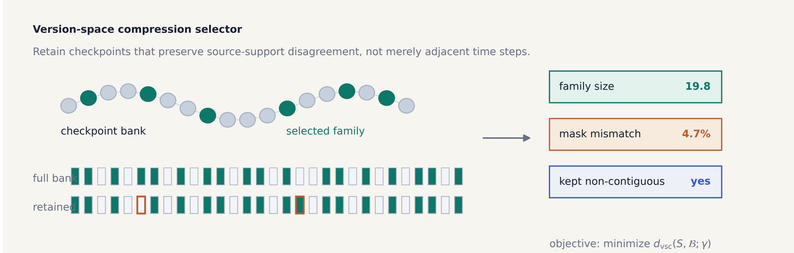
Step 2: Weight the retained family.

- CDA-Piv: use leave-one-out pivotality, but guard against checkpoints with high source-mixture sensitivity.
- CDA-BD: solve a simplex projection problem toward a balanced low-loss source target set.

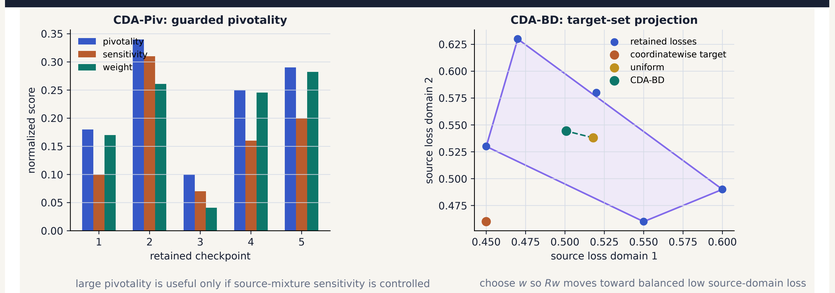
Step 3: Deploy one soup. The final model is

$$\bar{\theta}(w) = \sum_{j=1}^{k} w_j \theta_j.$$

Retained-Family Selection



Weighting Rules



Headline Results

Across the five-benchmark comparison package, CDA reaches

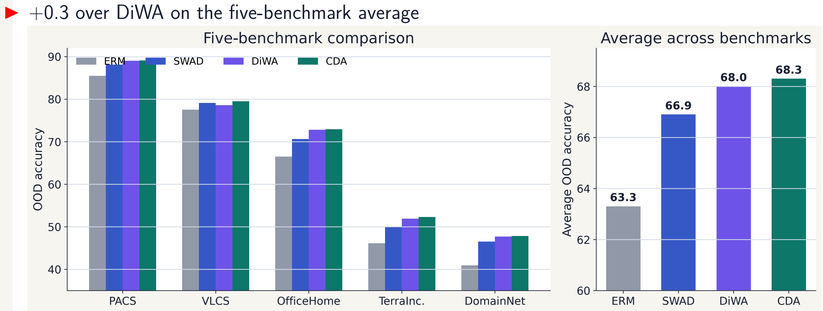
68.3

average OOD accuracy, compared with

66.9 for SWAD, 68.0 for DiWA, 63.3 for ERM.

Real headline reading

- +5.0 over ERM on the five-benchmark average
- +1.4 over SWAD on the five-benchmark average
- +0.3 over DiWA on the five-benchmark average

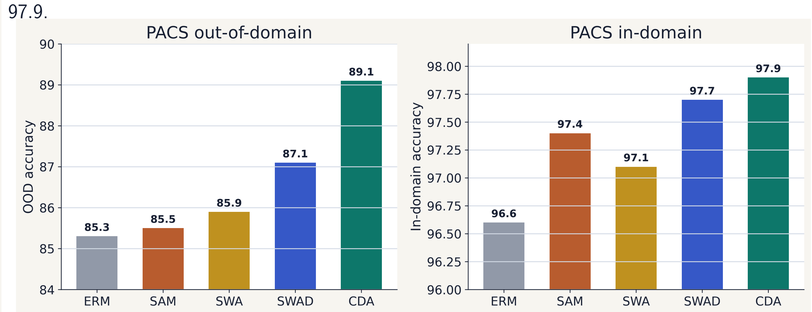


PACS SWAD-Style Comparison

On the PACS comparison aligned with the SWAD reporting format, CDA reaches:

89.1 OOD accuracy and 97.9 in-domain accuracy.

That improves the SWAD OOD number from 87.1 to 89.1 and the in-domain number from 97.7 to 97.9.



Takeaways

1. **Post-hoc DG can be principled.** The deployment choice can be governed by a source-only certificate rather than by time or heuristic averaging alone.
2. **Selection matters first.** The retained family must preserve the disagreement structure of the full checkpoint bank.
3. **Weighting matters second.** Once a good family is retained, CDA turns that structure into a better deployment decision through source-mixture-aware weighting.
4. **Averaging only helps when the soup is mergeable.** Flatness and local smoothness explain when endpoint-domain quantities transfer to the deployed average.

Limitations and Scope

- CDA studies a **post-hoc deployment problem**, not the full space of training-time DG algorithms.
- The theory assumes a **local source-supported hidden-shift model**. If the target lies far outside nearby source-mixture redistributions, the certificate can loosen.
- The strongest empirical support on this poster comes from the **settled benchmark tables** shown here.